Contrapositive closure of tableaux inconsistency

The Open Logic Project contributors

openlogic:first-order-logic/tableaux:local-snapshot-20261013; CC BY 4.0

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Open Logic Project unit file sources/olp-first-order-logic-tableaux/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the proof displays the two signed assumption sets and argues the Cut combination in prose, so no tableau environment occurs in the counted span

Editorial note. The label prop:provability-contr names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own prop environment in the unit source.

Allowed standard prerequisites (editorial).

- Inconsistency via closed tableaux. A set of sentences is inconsistent iff some finite subset has a closed tableau; the witnesses are closed tableaux for finite signed assumption sets built from the set.. Inconsistency of Gamma union {A} is unpacked into a finite witness C_1,...,C_m from Gamma with a closed tableau carrying True A among its assumptions.
- The Cut rule. A tableau branch may be split on any sentence A into a branch with True A and one with False A.. The two closed witnesses are combined into one closed tableau by a single Cut application on A.

Proposition 0.1. *If* $\Gamma \vdash A$ *and* $\Gamma \cup \{A\}$ *is inconsistent, then* Γ *is inconsistent.*

Proof. There are finite $\Gamma_0 = \{B_1, \ldots, B_n\}$ and $\Gamma_1 = \{C_1, \ldots, C_n\} \subseteq \Gamma$ such that

$$\{\bot(A), \top(B_1), \ldots, \top(B_n)\}$$
$$\{\top(A), \top(C_1), \ldots, \top(C_m)\}$$

have closed tableaus. Using the Cut rule on A we can combine these into a single closed tableau that shows $\Gamma_0 \cup \Gamma_1$ is inconsistent. Since $\Gamma_0 \subseteq \Gamma$ and $\Gamma_1 \subseteq \Gamma$, $\Gamma_0 \cup \Gamma_1 \subseteq \Gamma$, hence Γ is inconsistent. □